CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

December 2011

EE 301 Electrical Machines - II

Date: 12.12.2011, Monday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 Draw and explain the torque speed (slip) characteristics of three phase induction motor and also explain the effect of rotor resistance on torque slip characteristics. **[07]**

Q – 2 Draw the circle diagram for a 5.6 kW, 400 V, three phase, 4 pole, 50 Hz slip ring induction motor from the following data: **[14]**
No load readings : 400 V, 6 A, $\cos\O_0 = 0.087$
Short circuit readings: 100 V, 12 A, 720 W.
The ratio of primary to secondary turns =2.62, Stator resistance per phase is 0.67 Ω and of the rotor is 0.185 Ω. Calculate (i) full load current (ii) full load slip (iii)full load power factor (iv) ratio of maximum torque to full load torque and (v)maximum power.

OR

Q - 2 (a) Explain harmonic induction torques and harmonic synchronous torques. **[07]**

(b) Explain in brief various methods of speed control of three phase induction motor. **[07]**

Q - 3 (a) The equivalent impedance of the outer and the inner cage rotors of a double cage induction motor are (0.8 + j0.4) Ω and (0.2 + j3.0) Ω, respectively. Find the ratio of torques produced by the two cages during (a) starting and (b) at full load. Full load slip is 4 %. **[07]**

(b) A three phase, four pole, 50 Hz, star connected induction motor is connected to a 400 V supply. The rotor resistance and leakage reactance are 0.2 Ω and 0.81 Ω, respectively. The ratio of stator to rotor effective turns is 1.8. If the full load slip is 4 %, calculate (a) the full load torque, (b) the full load output, (c) the maximum torque, (d) the speed at maximum torque, and (e) the starting torque. **[07]**

OR

Q - 3 (a) Explain the principle, working and application of magnetic levitation. **[05]**

(b) Explain the double cage motor and draw its equivalent circuit. **[05]**

(c) Draw the power flow diagram of induction motor and explain the power stages. Draw the phasor diagram. **[04]**

SECTION – II

Q - 4 Explain principle, working and application of induction generator. Draw the characteristics of induction generator showing the effect of capacitor. [07]

Q - 5 (a) State the different types of single phase induction motor. Explain in detail construction, working, starting and running performance, characteristics and application of any one type of single phase induction motor. [07]

(b) Give classification of stepper motor and explain any one of them in detail. [07]

OR

Q - 5 (a) Discuss the procedure for determining the parameters of equivalent circuit of single phase induction motors. Also draw an equivalent circuit for it. [07]

(b) A 250 W, 230 V, 50 Hz capacitor start motor has the following constants for the main and auxiliary windings: Main winding, $Z_m = (4.5 + j\,3.7)\ \Omega$. Auxiliary winding $Z_a = (9.5 + j3.5)\ \Omega$. Determine the value of the starting capacitor that will place the main and auxiliary winding currents in quadrature at starting. [07]

Q - 6 (a) Explain with neat sketch the construction and working of Schrage Motor. Describe how it can control speed and power factor. [07]

(b) A squirrel cage type induction motor when started by means of a star/delta starter takes 180% of full load line current and develops 35% of full load torque at starting. Calculate the starting torque and current in terms of full load values, if an auto transformer with 75% tapping were employed. [07]

OR

Q - 6 (a) Explain in brief shaded pole induction motor. [04]

(b) Write a note on universal motor and its application. [05]

(c) Write a note on construction working and application of repulsion motor. [05]

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